

Find the sum of all integer bases $b > 9$ for which 17_b is a divisor of 97_b .

Model Output



We are given the problem: Find the sum of all integer bases [... 569 tokens elided] Step 4: Check the values of b We need to check that these values of b satisfy the original condition: $17_b \mid 97_b$ - For $b = 21$: $17_{21} = 1 \cdot 21 + 7 = 28$; $97_{21} = 9 \cdot 21 + 7 = 196$ Check if $28 \mid 196$: Yes, $196 \div 28 = 7$ - For $b = 49$: $17_{49} = 1 \cdot 49 + 7 = 56$; $97_{49} = 9 \cdot 49 + 7 = 443$ Check if $56 \mid 443$: $443 \div 56 = 7.91\dots$, so **not** an integer. Wait — this contradicts our earlier logic. Let's double-check. We had: $(9b + 7)/(b + 7) = 9 - 56/(b + 7)$ So for this to be an integer, $56/(b + 7)$ must be an integer, i.e., $b + 7 \mid 56$ So $b + 7 \in \{1, 2, 4, 7, 8, 14, 28, 56\}$ But we want $b > 9 \Rightarrow b + 7 > 16$ So only $b + 7 = 28, 56$ are valid. Then: - $b = 21$: valid - $b = 49$: valid But earlier, when we checked $b = 49$, we saw that $97_{49} = 443$, and $17_{49} = 56$, and $443 \div 56 = 7.91$, not an integer. So $b = 49$ is **not valid**. But wait — that contradicts the earlier logic. What's the issue? Let's recompute 97_{49} : $97_{49} = 9 \cdot 49 + 7 = 441 + 7 = 448$ Ah! That's the correct value. Then: $448 \div 56 = 8$ So $56 \mid 448$, so $b = 49$ is valid. So both $b = 21$ and $b = 49$ are valid. Final Step: Sum the valid bases $21 + 49 = 70$ Final Answer: **70**

